



CLOUDCART ONLINE STORE

Prepared by :

Esraa Samir

Lobna Ashraf

Hager Mahmoud

Supervised by

Eng. Mostafa Ashraf

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Chapter 1

Introduction

This chapter provides a brief introduction to the CloudCart project and its AWS cloud architecture, highlighting the main goals and services used.

1.1 Project Overview

This project presents the design and implementation of a highly available and scalable cloud-based architecture for CloudCart Online Store using Amazon Web Services (AWS).

The architecture is designed to support an online shopping application by providing secure networking, scalable compute resources, reliable storage, database services, and monitoring capabilities. The solution follows AWS best practices to improve availability, scalability, security, and performance.

The implemented architecture uses multiple AWS services, including Amazon VPC, EC2, Application Load Balancer, Auto Scaling, Amazon RDS, Amazon S3, Amazon EFS, AWS Lambda, IAM, Amazon CloudWatch, and Amazon SNS.

1.2 Project Objectives

The main objectives of the project are to:

- Design a secure and scalable AWS cloud infrastructure.
- Deploy the application across multiple Availability Zones to improve availability.
- Separate public, private application, and private database resources.
- Implement load balancing and Auto Scaling for improved application performance and scalability.
- Store application data using appropriate AWS storage services.
- Deploy a private and highly available database using Amazon RDS.
- Implement monitoring and notification mechanisms.
- Apply security best practices using IAM and Security Groups.

1.3 Documentation Purpose

This document provides a step-by-step overview of the implementation of the CloudCart AWS architecture.

It documents the configuration of the main AWS services used in the project and provides screenshots from the AWS Management Console to demonstrate the implemented configurations. Each section briefly explains the purpose of the configured service and its role within the overall architecture.

1.4 AWS Architecture Overview

The CloudCart architecture is built within an Amazon VPC spanning two Availability Zones. It includes public subnets for internet-facing components, private application subnets for the EC2 instances, and private database subnets for Amazon RDS.

An Application Load Balancer distributes incoming traffic across the application servers, while Auto Scaling helps maintain the required number of EC2 instances. Amazon EFS provides shared file storage, Amazon S3 is used for storing product images and documents, and AWS Lambda processes S3 upload events.

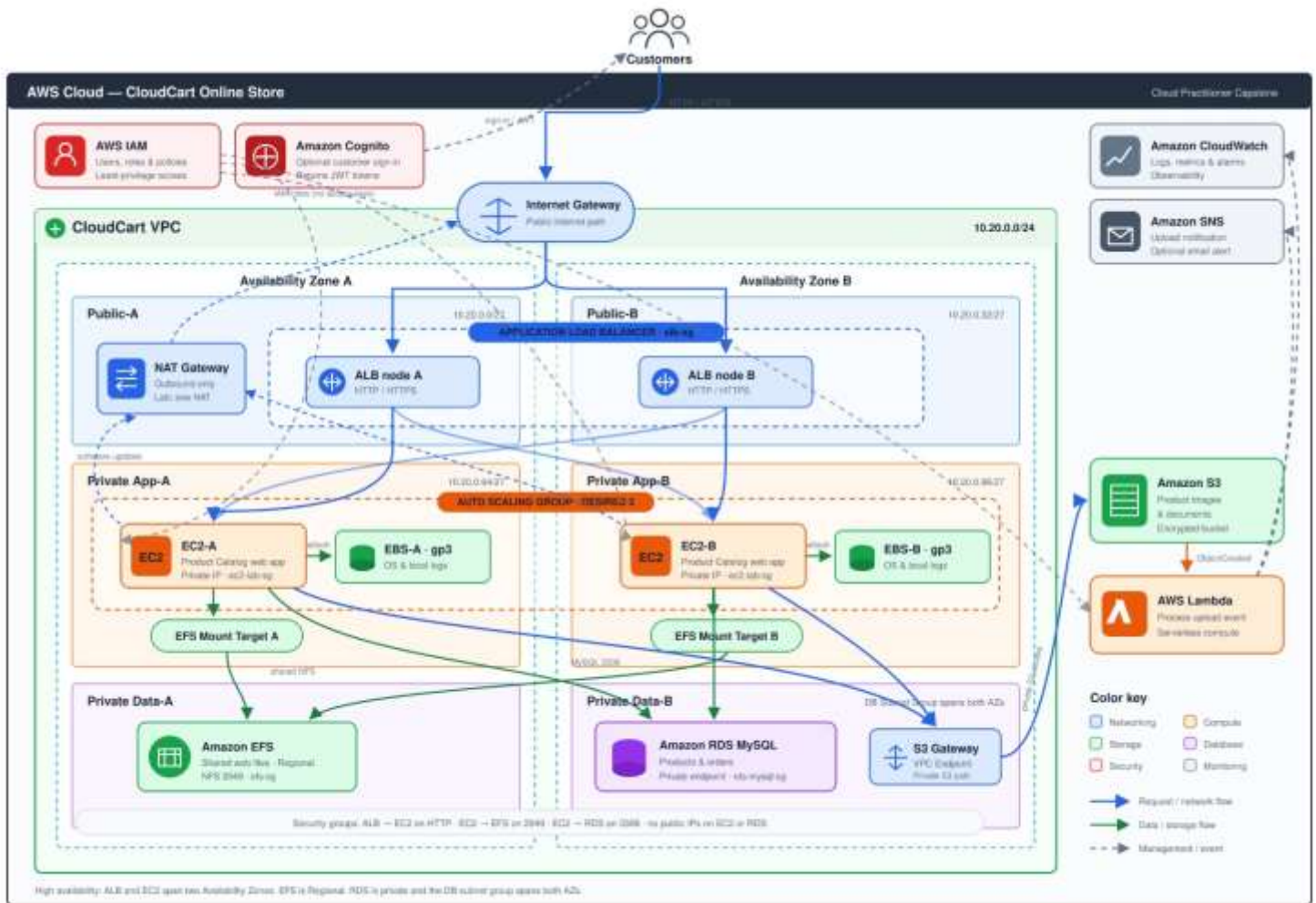
The architecture also includes monitoring and notification services such as Amazon CloudWatch and Amazon SNS to improve observability and operational management.

Chapter 2

System Architecture

This chapter presents the overall AWS architecture of the CloudCart project and shows how the main AWS services are connected to support the application.

2.1 Architecture Diagram



The architecture illustrates the main AWS services used in the CloudCart application and how they interact within the AWS environment. The design includes networking, compute, storage, database, security, and monitoring components distributed across multiple Availability Zones to provide scalability, availability, and security.

2.2 AWS Architecture Overview

The CloudCart architecture consists of several AWS services working together to provide a secure, scalable, and highly available cloud environment.

The main components include:

- Networking: Amazon VPC, Subnets, Internet Gateway, NAT Gateway, and Route Tables.
- Compute: Amazon EC2, Auto Scaling, and Application Load Balancer.
- Storage: Amazon S3, Amazon EFS, and Amazon EBS.
- Database: Amazon RDS for MySQL.
- Serverless: AWS Lambda and S3 Gateway Endpoint.
- Security: IAM, Security Groups, and Amazon Cognito.
- Monitoring: Amazon CloudWatch and Amazon SNS.

2.3 Architecture Flow

User requests are received through the Application Load Balancer and distributed across the EC2 instances in the private application subnets. The application interacts with Amazon RDS for database operations, while Amazon S3 and Amazon EFS provide storage services. AWS Lambda handles S3-related events, and Amazon CloudWatch monitors the environment with Amazon SNS providing notifications.

Chapter 3

AWS Implementation

This chapter documents the implementation and configuration of the main AWS services used in the CloudCart project.

3.1 VPC Configuration



The CloudCart VPC was created to provide an isolated and secure networking environment for the application. It uses the IPv4 CIDR block "192.168.1.0/24" and is configured with DNS resolution enabled. The VPC serves as the main networking layer for the application, connecting the required public and private resources across multiple Availability Zones.



The CloudCart VPC was created to provide an isolated and secure networking environment for the application. It uses the IPv4 CIDR block "192.168.1.0/24" and is configured with DNS resolution enabled. The VPC serves as the main networking layer for the application, connecting the required public and private resources across multiple Availability Zones.

3.2 Subnets Configuration

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Public-B-ALB	subnet-04f44d893b305d1ee	Available	vpc-09e19e479da163d21 CloudCart VPC	Off	192.168.1.32/27
Public-A-ALB	subnet-0247fad95075e3cd8	Available	vpc-09e19e479da163d21 CloudCart VPC	Off	192.168.1.0/27
Private-Data-A	subnet-0f32716c360822a31	Available	vpc-09e19e479da163d21 CloudCart VPC	Off	192.168.1.128/2
Private-App-B	subnet-0ed662e6e770b0f5e	Available	vpc-09e19e479da163d21 CloudCart VPC	Off	192.168.1.192/2
Private-App-A	subnet-055065983e3218d80	Available	vpc-09e19e479da163d21 CloudCart VPC	Off	192.168.1.64/27
Private-Data-B	subnet-0d30d8e1700a2c9af	Available	vpc-09e19e479da163d21 CloudCart VPC	Off	192.168.1.160/2

To ensure high availability, fault tolerance, and proper isolation of resources, the CloudCart VPC is segmented into multiple public and private subnets distributed across two Availability Zones (AZ-A and AZ-B).

- **Public Subnets (Public-A-ALB & Public-B-ALB):** Configured to handle internet-facing traffic and host the Application Load Balancer (ALB).
- **Private App Subnets (Private-App-A & Private-App-B):** Dedicated to hosting the backend compute instances (EC2) securely without direct exposure to the public internet.
- **Private Data Subnets (Private-Data-A & Private-Data-B):** Isolated subnets reserved for database and storage tiers (such as Amazon RDS and EFS) to maintain maximum security and data privacy.

subnet-04f44d893b305d1ee / Public-B-ALB

Details

Subnet ID
subnet-04f44d893b305d1ee

IPv4 CIDR
192.168.1.32/27

Availability Zone
us-east-1b (us-east-1b)

Network ACL
acl-0fa1ca3238b51427b

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-east-1:294427322944:subnet/subnet-04f44d893b305d1ee

Available IPv4 addresses
26

Network border group
us-east-1

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
-

VPC
vpc-09e19e479da163d21 | CloudCart VPC

Auto-assign public IPv4 address
No

Outpost ID
-

Hostname type
IP name

Owner
294427322944

Block Public Access
Off

IPv6 CIDR association ID
-

Route table
rtb-0e65ac9b0e4b73cd3d | CloudCart Public RouteTable

Auto-assign IPv6 address
No

IPv4 CIDR reservations
-

Resource name DNS A record
Disabled

The **Public-B-ALB** subnet (subnet-04f44d893b305d1ee) is deployed in Availability Zone use1-az2 (us-east-1b) with the IPv4 CIDR block 192.168.1.32/27. It is associated with the public route table to handle internet-facing traffic and hosts the

Application Load Balancer (ALB) node in this zone, ensuring high availability and public accessibility.

subnet-0247fad95076e3dd8 / Public-A-ALB			
Details			
Subnet ID subnet-0247fad95076e3dd8	Subnet ARN arn:aws:ec2:us-east-1:294427322944:subnet/subnet-0247fad95076e3dd8	State Available	Block Public Access Off
IPv4 CIDR 192.168.1.0/27	Available IPv4 addresses 26	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1a (us-east-1a)	Network border group us-east-1	VPC vpc-09e19e479da163d21 CloudCart-VPC	Route table rtb-0069acfbba4b73d3d CloudCart-Public-RouteTable
Network ACL acl-0fa1ca3238b51427e	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 294427322944	

The **Public-A-ALB** subnet (subnet-0247fad95076e3dd8) is deployed in Availability Zone use1-az1 (us-east-1a) with the IPv4 CIDR block 192.168.1.0/27. It is linked to the public route table to route external client traffic and host the Application Load Balancer node in this zone, supporting high availability and traffic distribution.

subnet-0f32716c360822a31 / Private-Data-A			
Details			
Subnet ID subnet-0f32716c360822a31	Subnet ARN arn:aws:ec2:us-east-1:294427322944:subnet/subnet-0f32716c360822a31	State Available	Block Public Access Off
IPv4 CIDR 192.168.1.128/27	Available IPv4 addresses 27	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1a (us-east-1a)	Network border group us-east-1	VPC vpc-09e19e479da163d21 CloudCart-VPC	Route table rtb-00af4ec779cd5ce88 Private-Data-RouteTable
Network ACL acl-0fa1ca3238b51427e	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 294427322944	

The **Private-Data-A** subnet (subnet-0f32716c360822a31) is deployed in Availability Zone use1-az1 (us-east-1a) with the IPv4 CIDR block 192.168.1.128/27. It is isolated from the public internet and associated with a private data route table, serving as a secure zone to host backend data storage tiers such as Amazon RDS MySQL.

subnet-0edb62c6e770b0f8e / Private-App-B

Actions

Details

Subnet ID

subnet-0edb62c6e770b0f8e

IPv4 CIDR

192.168.1.192/27

Availability Zone

use1-az2 (us-east-1b)

Network ACL

acl-0fa1ca3238b51427e

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:294427322944:subnet/subnet-0edb62c6e770b0f8e

Available IPv4 addresses

24

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv4 CIDR

-

VPC

vpc-09e19e479da163d21 | CloudCart-VPC

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

294427322944

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-068af4f3707724838 | CloudCart-Nat-Root-Table

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-09e19e479da163d21 | CloudCart-VPC

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

294427322944

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-058af4f3707724838 | CloudCart-Nat-Root-Table

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

The **Private-App-B** subnet (subnet-0edb62c6e770b0f8e) is deployed in Availability Zone use1-az2 (us-east-1b) with the IPv4 CIDR block 192.168.1.192/27. It is completely isolated from direct public traffic, linked to a private route table routed through the NAT gateway, and designed to securely host backend application instances (EC2).

subnet-055065983e32f8d80 / Private-App-A

Actions

Details

Subnet ID

subnet-055065983e32f8d80

IPv4 CIDR

192.168.1.64/27

Availability Zone

use1-az1 (us-east-1a)

Network ACL

acl-0fa1ca3238b51427e

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:294427322944:subnet/subnet-055065983e32f8d80

Available IPv4 addresses

24

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-09e19e479da163d21 | CloudCart-VPC

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

294427322944

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-098af4f3707724838 | CloudCart-Net-Route-Table

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-09e19e479da163d21 | CloudCart-VPC

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

294427322944

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-058af4f3707724838 | CloudCart-Nat-Root-Table

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

The **Private-App-A** subnet (subnet-055065983e32f8d80) is deployed in Availability Zone use1-az1 (us-east-1a) with the IPv4 CIDR block 192.168.1.64/27. It is kept secure from the public internet, associated with the private route table routed through the NAT gateway, and used to host the primary backend application instances (EC2).

subnet-0d30d8e1700a2c9af / Private-Data-B			
Actions			
Details			
Subnet ID subnet-0d30d8e1700a2c9af	Subnet ARN arn:aws:ec2:us-east-1:294427322944:subnet/subnet-0d30d8e1700a2c9af	State Available	Block Public Access Off
IPv4 CIDR 192.168.1.160/27	Available IPv4 addresses 26	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1b (us-east-1b)	Network border group us-east-1	VPC vpc-09e19e479da163d21 CloudCart-VPC	Route table rtb-00af4ec779cd5ce98 Private-Data-RouteTable
Network ACL acl-0fa1ca3238b51427*	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 294427322944	

The **Private-Data-B** subnet (subnet-0d30d8e1700a2c9af) is deployed in Availability Zone `us-east-1b` with the IPv4 CIDR block `192.168.1.160/27`. It is completely isolated from the public internet, associated with a private data route table, and functions alongside Private-Data-A to securely host backend storage and database instances like Amazon RDS MySQL across multiple availability zones.

3.3 Internet Gateway (IGW) Configuration

igw-05c1582815e0e9a16 / public-CloudCart-IGW			
Actions			
Details			
Internet gateway ID igw-05c1582815e0e9a16	State Attached	VPC ID vpc-09e19e479da163d21 CloudCart-VPC	Owner 294427322944

The **Internet Gateway (IGW)** (igw-05c1582815e0e9a16), named `public-CloudCart-IGW`, is attached to the CloudCart VPC (vpc-09e19e479da163d21). It serves as the vital component that enables communication between the VPC and the public internet, allowing internet-facing traffic to reach the public subnets and the Application Load Balancer.

3.4 NAT Gateway Configuration

nat-12168370d9ab5b240 / private_CloudCart-NAT

Details

- NAT gateway ID: nat-12168370d9ab5b240
- NAT gateway ARN: arn:aws:ec2:us-east-1:294427322944:natgateway/nat-12168370d9ab5b240
- VPC: vpc-09e19e479da163d21 / CloudCart-VPC
- Availability mode: Regional
- Connectivity type: Public
- Method of EIP allocation: Automatic
- State: Available
- Created: Tuesday, August 25, 2026 at 11:39:13 GMT+5
- Deleted: -

Associated IP addresses (2)

IP address	Status	Availability Zone	Allocation ID
34.199.196.170	Succeeded	us-east-1a (us-east-1a)	eipalloc-02c0912ce458b996c
18.200.224.192	Succeeded	us-east-1b (us-east-1b)	eipalloc-025c248009fec7cb5

The **NAT Gateway** (nat-12168370d9ab5b240), named private_CloudCart-NAT, is deployed within the CloudCart VPC to enable instances in the private subnets (such as backend application servers) to initiate outbound internet communication (e.g., for software updates or patches) while preventing any unsolicited inbound traffic from the public internet. It is configured with public connectivity type and associated with Elastic IP addresses across availability zones.

3.5 Route Tables Configuration

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	Private-Data-RouteTable	rtb-00af4ec729cd5ce98	2 subnets	-	No	vpc-09e19e479da163d21 Clo..
☐	CloudCart-Public-RootTable	rtb-0869acfbba4b7d3d	2 subnets	-	No	vpc-09e19e479da163d21 Clo..
☐	CloudCart-Main-RootTable	rtb-09ae11e9ae50c5ca	-	-	Yes	vpc-09e19e479da163d21 Clo..
☐	AutoScaling-Private-Route-Table	rtb-0e0a2e19bc038f185	-	nat-12168370d9ab5b240	No	vpc-09e19e479da163d21 Clo..
☐	CloudCart-Nat-Root-Table	rtb-098af4f370724838	2 subnets	-	No	vpc-09e19e479da163d21 Clo..

The **Route Tables** define the rules (routes) that dictate where network traffic is directed from the subnets within the CloudCart VPC (vpc-09e19e479da163d21). The configuration includes the main route table, public route tables (CloudCart-Public-RootTable) connected to the Internet Gateway, and private/NAT route tables (CloudCart-Nat-RootTable, Private-Data-RouteTable) routing traffic securely through the NAT Gateway.

3.6 Security Groups Configuration

☐	Name	Security group ID	Security group name	VPC ID
☐	CloudCart-RDS-SG	sg-002a82a5781eaf151	CloudCart-RDS-SG	vpc-09e19e479da163d21
☐	CloudCart-EFS-SG	sg-0a284fc70d00e7000	CloudCart-EFS-SG	vpc-09e19e479da163d21
☐	CloudCart-EC2-SG	sg-02e6ff0fdc04c308e	CloudCart-EC2-SG	vpc-09e19e479da163d21
☐	-	sg-06f5a88773ede84e2	default	vpc-09e19e479da163d21
☐	CloudCart-ALB-SG	sg-06eb9fd2d793e29fd	CloudCart-ALB-SG	vpc-09e19e479da163d21

Description	Owner	Inbound rules count	Outbound rules count
Allow MySQL access from CloudCart EC...	294427322944	1 Permission entry	1 Permission entry
Allow NFS access from CloudCart EC2 in...	294427322944	1 Permission entry	1 Permission entry
Allow HTTP from ALB only	294427322944	1 Permission entry	1 Permission entry
default VPC security group	294427322944	1 Permission entry	1 Permission entry
Allow HTTP from Internet to ALB	294427322944	1 Permission entry	1 Permission entry

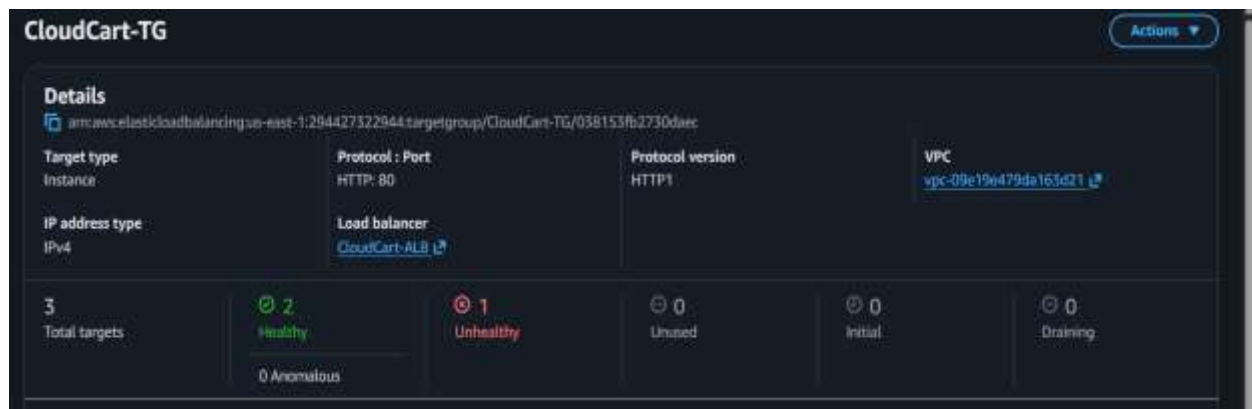
The **Security Groups** function as virtual stateful firewalls controlling inbound and outbound traffic for resources within the CloudCart VPC ([vpc-09e19e479da163d21](#)). Dedicated security groups are implemented with strict least-privilege rules, including **CloudCart-ALB-SG** for public HTTP traffic, **CloudCart-EC2-SG** allowing traffic exclusively from the ALB, and internal security groups (**CloudCart-EFS-SG** and **CloudCart-RDS-SG**) restricted solely to application server access.

3.7 Application Load Balancer (ALB) Configuration

CloudCart-ALB			
Details			
Load balancer type Application	Status Active	VPC vpc-09e19e479da163d21	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35XDOTRQ7X7K	Availability Zones subnet-04f44a893b305d1ee us-east-1b (use1-az2) subnet-0247fad95076e3dd8 us-east-1a (use1-az1)	Date created August 25, 2026, 19:28 (UTC+03:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:294427322944:loadbalancer/app/CloudCart-ALB/d3616f8e5bdc26bb		DNS name info CloudCart-ALB-1391285923.us-east-1.elb.amazonaws.com (A Record)	

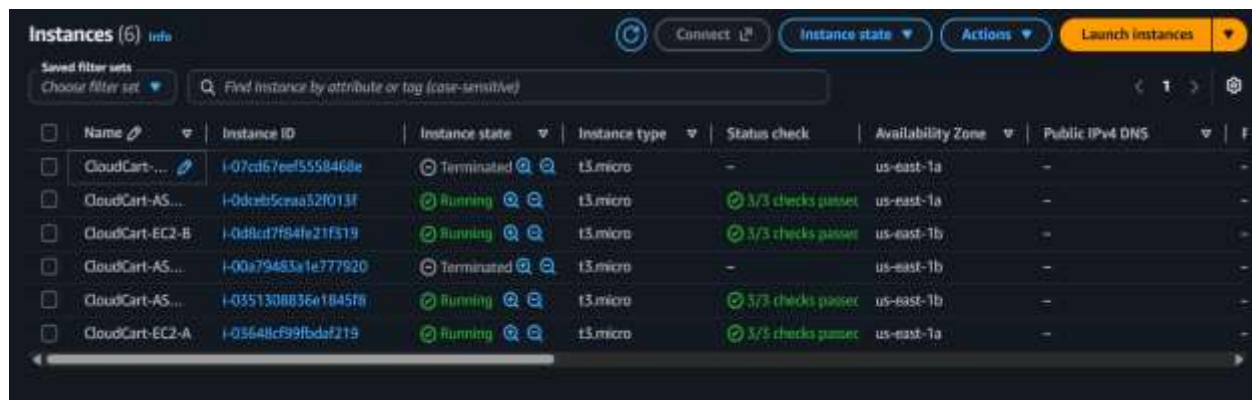
The **Application Load Balancer (ALB)** (CloudCart-ALB) is deployed as an internet-facing load balancer across multiple Availability Zones (use1-az1 and use1-az2) within the public subnets of the CloudCart VPC. It acts as the single point of

entry for incoming customer HTTP/HTTPS traffic, efficiently distributing requests to the backend application instances.



The **Target Group** (CloudCart-TG) is configured to route incoming requests from the Application Load Balancer to backend targets on port 80 using the Instance target type within the CloudCart VPC. It continuously monitors target health (showing healthy EC2 instances) to ensure high availability and reliable traffic distribution.

3.9 Amazon EC2 Instances Configuration



The **Amazon EC2 Instances** serve as the core compute layer for the CloudCart application, deployed across multiple Availability Zones (us-east-1a and us-east-1b) within the private subnets. Utilizing t3.micro instances, the architecture supports both standalone instances and dynamic instances managed via Auto Scaling, ensuring high availability, fault tolerance, and seamless integration with the Application Load Balancer.

Instance summary for i-03648cf99fbdaf219 (CloudCart-EC2-A)

Updated less than a minute ago

[Connect](#)
[Instance state](#)
[Actions](#)

Instance ID i-03648cf99fbdaf219	Public IPv4 address -	Private IPv4 addresses 192.168.1.87
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-192-168-1-87.ec2.internal	Private IP DNS name (IPv4 only) ip-192-168-1-87.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation Learn more
Auto-assigned IP address -	VPC ID vpc-09e19e479da163d21 (CloudCart-VPC)	Auto Scaling Group name -
IAM role CloudCart-EC2-SSM-Role	Subnet ID subnet-055065963e32f8d80 (Private-App-A)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:294427322944:instance/i-03648cf99fbdaf219	
Operator -		

Instance summary for i-0d8cd7f84fe21f319 (CloudCart-EC2-B)

Updated less than a minute ago

[Connect](#)
[Instance state](#)
[Actions](#)

Instance ID i-0d8cd7f84fe21f319	Public IPv4 address -	Private IPv4 addresses 192.168.1.218
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-192-168-1-218.ec2.internal	Private IP DNS name (IPv4 only) ip-192-168-1-218.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation Learn more
Auto-assigned IP address -	VPC ID vpc-09e19e479da163d21 (CloudCart-VPC)	Auto Scaling Group name -
IAM role CloudCart-EC2-SSM-Role	Subnet ID subnet-0ed62c5e770bd0f8e (Private-App-B)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:294427322944:instance/i-0d8cd7f84fe21f319	
Operator -		

3.10 Auto Scaling Configuration

The Auto Scaling Group (CloudCart-ASG) is configured to maintain application availability and elasticity, with a desired capacity of 2 instances and scaling limits ranging between 2 and 4 instances. Utilizing the CloudCart-ASG-Template launch template with t3.micro instances, the group dynamically adjusts compute capacity based on traffic demands across the designated availability zones.

CloudCart-ASG Capacity overview

Edit

arn:aws:autoscaling:us-east-1:294427322944:autoScalingGroup:7ec2a8f-275c-4178-9d9b-9b0e25bddbc0:autoScalingGroupName/CloudCart-ASG

Desired capacity

2

Scaling limits

2 - 4

Desired capacity type

Units (number of instances)

Status

At desired capacity

Date created

Tue Aug 25 2026 20:32:34 GMT+0300 (Eastern European Summer Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template

it-0568b5a36862553tf2

CloudCart-ASG-Template

Version

Default

Description

install apache

AMI ID

ami-0332d564d76dbd8d5

Security groups

-

Storage (volumes)

-

Instance type

t3.micro

Security group IDs

sg-02e6ff0f0dc04c308e

Key pair name

Final-CloudCart-EC2-A

Owner

arn:aws:iam:294427322944:root

Create time

Tue Aug 25 2026 20:41:43 GMT+0300 (Eastern European Summer Time)

Request Spot instances

No

3.11 Amazon EBS (Elastic Block Store) Configuration

vol-0693cb6e4441af7f1 (CloudCart-EBS-A)

Last updated

less than a minute ago

Actions

Delete

Modify

Details

Volume ID

vol-0693cb6e4441af7f1 (CloudCart-EBS-A)

Size

10 GiB

Type

gp3

Status check

Ok

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Volume state

In-use

IOPS

3000

Throughput

125

Fast snapshot restored

No

Availability Zone

use1-az1 (us-east-1a)

Created

Tue Aug 25 2026 21:15:11 GMT+0300 (Eastern European Summer Time)

Multi-Attach enabled

No

Attached resources

i-03648cf99fbdaf219 (CloudCart-EC2-A) (EBS card index 0): /dev/sdf (attached)

Outposts ARN

-

Managed

false

Operator

-

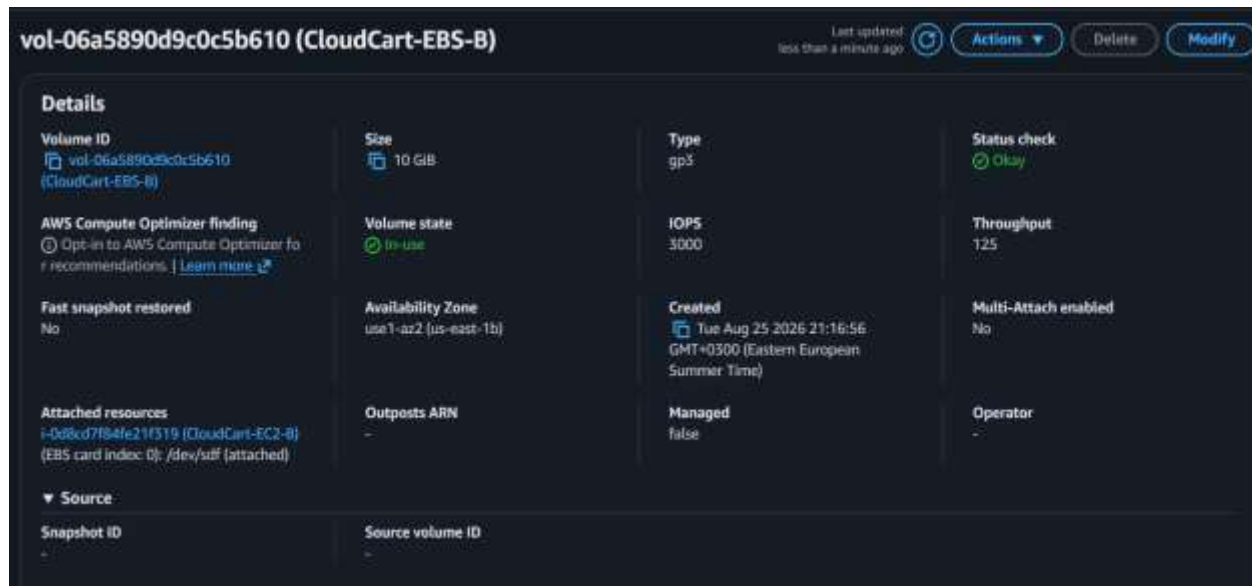
Source

Snapshot ID

-

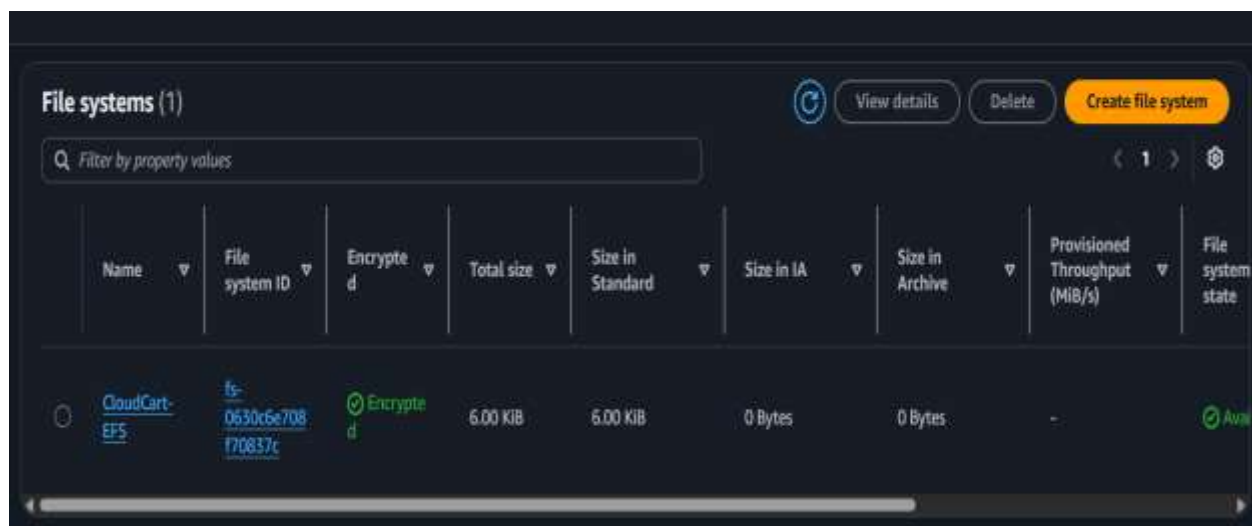
Source volume ID

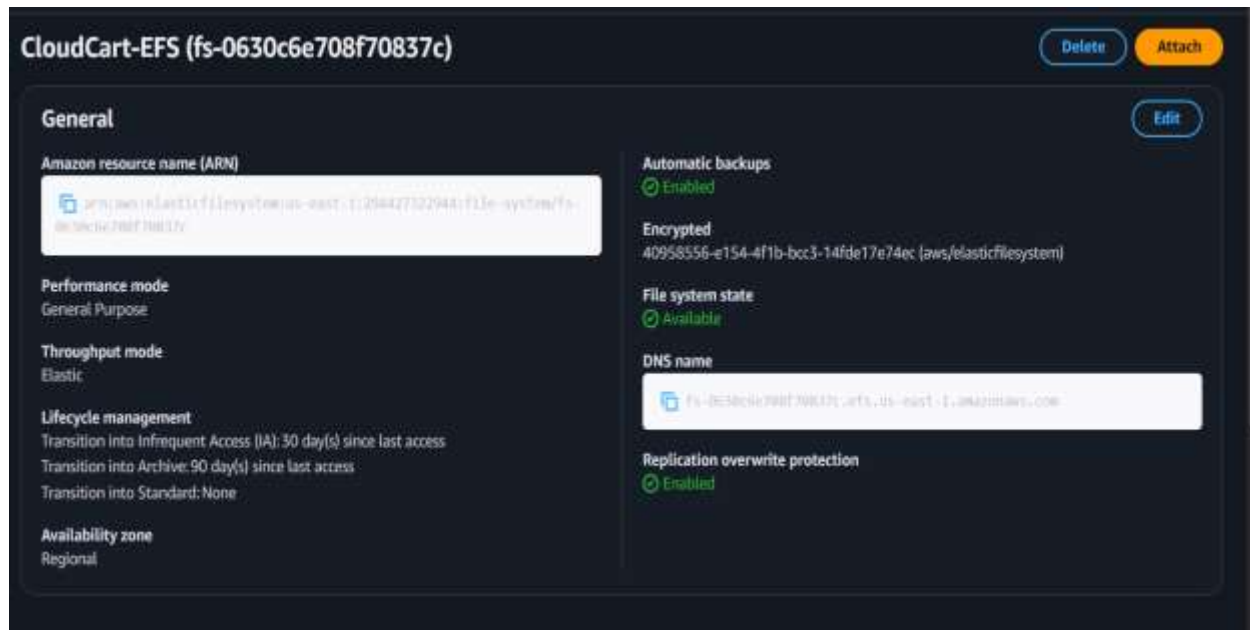
-



Amazon **Elastic Block Store (EBS)** volumes (gp3, 10 GiB) are provisioned and attached to the respective EC2 instances across different Availability Zones (us-east-1a and us-east-1b) within the CloudCart environment. These persistent block storage volumes ensure high performance with dedicated IOPS and seamless attachment to the primary application servers.

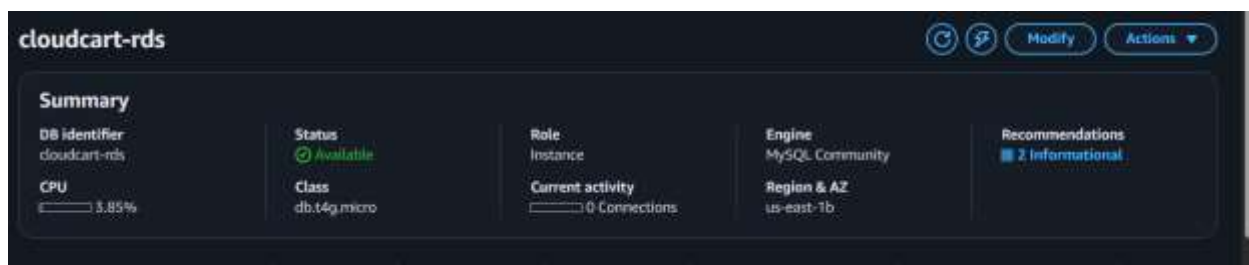
3.12 Amazon EFS (Elastic File System) Configuration





Amazon **Elastic File System (EFS)** (CloudCart-EFS) is integrated into the architecture to provide fully managed, scalable, and shared file storage across multiple availability zones. Configured with encryption, automatic backups, and optimized performance modes, it allows backend EC2 instances to securely share application data seamlessly.

3.13 Amazon RDS (Relational Database Service) Configuration



The **Amazon RDS** instance (cloudcart-rds) is configured using the **MySQL Community** engine with a db.t4g.micro instance class. Operating in the us-east-1b availability zone, the database is fully active and available, providing a secure and reliable persistent backend storage layer for the CloudCart application.

3.14 Amazon S3 (Simple Storage Service) Configuration

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (3) [Info](#)

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

Name	AWS Region	Creation date
cloudcart-esraa-files-2026	US East (N. Virginia) us-east-1	August 26, 2026, 20:36:31 (UTC+03:00)

Account snapshot [Info](#)

Updated daily

[View dashboard](#)

Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)

cloudcart-esraa-files-2026 [Info](#)

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name	Type	Last modified	Size	Storage class
se (1).sql	sql	August 27, 2026, 17:13:24 (UTC+03:00)	11.1 KB	Standard
SE-project1.zip	zip	August 27, 2026, 15:55:39 (UTC+03:00)	661.1 KB	Standard

Amazon **S3 (Simple Storage Service)** (cloudcart-esraa-files-2026) is utilized to securely store static assets, database backups, and project files. Configured in the us-east-1 region, it provides a highly durable, scalable, and secure object storage layer for the CloudCart application.

3.15 AWS Lambda Configuration



The **AWS Lambda** function (CloudCart-S3-Processor) is implemented as a serverless component configured with an **S3 trigger** to process uploaded files automatically. Configured with standard execution parameters (128 MB memory and a 3-second timeout), it executes serverless data processing tasks efficiently without managing underlying infrastructure.

3.16 S3 Gateway Endpoint Configuration



A **VPC Gateway Endpoint** (CloudCart-S3-Endpoint) is configured for Amazon S3 (com.amazonaws.us-east-1.s3) within the CloudCart-VPC. This allows private and secure communication between resources inside the VPC and Amazon S3 without requiring internet gateways or NAT devices, enhancing security and reducing data transfer costs.

3.17 AWS IAM (Identity and Access Management) Configuration

The first screenshot shows the 'CloudCart-EC2-SSM-Role' page. It includes a summary section with the following details:

- Creation date:** August 25, 2026, 19:17 (UTC+03:00)
- ARN:** `arn:aws:iam::294427322944:role/CloudCart-EC2-SSM-Role`
- Instance profile ARN:** `arn:aws:iam::294427322944:instance-profile/CloudCart-EC2-SSM-Role`
- Last activity:** 10 minutes ago
- Maximum session duration:** 1 hour

The 'Permissions policies (2)' section shows two attached policies:

Policy name	Type	Attached entities
AmazonSSMManagedInstanceCore	AWS managed	1
SSM-SS	Customer inline	0

The second screenshot shows the 'CloudCart-S3-Processor-role-vblgaah0' page. It includes a summary section with the following details:

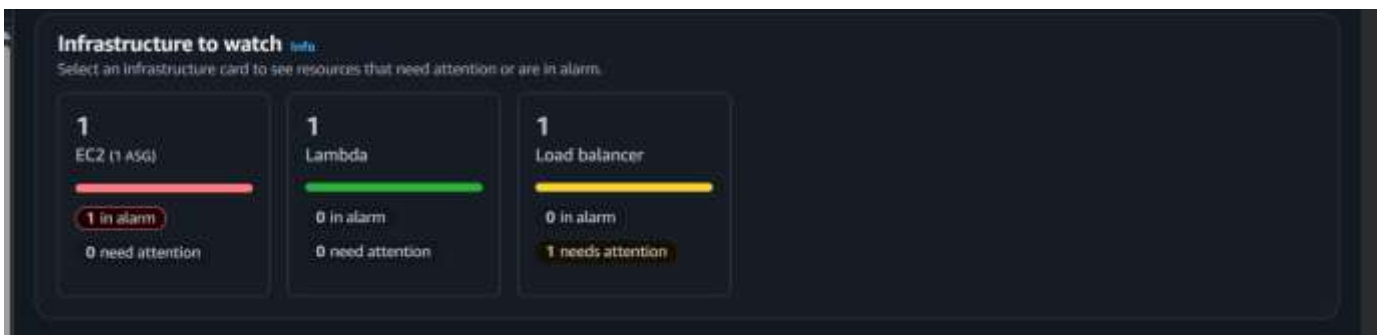
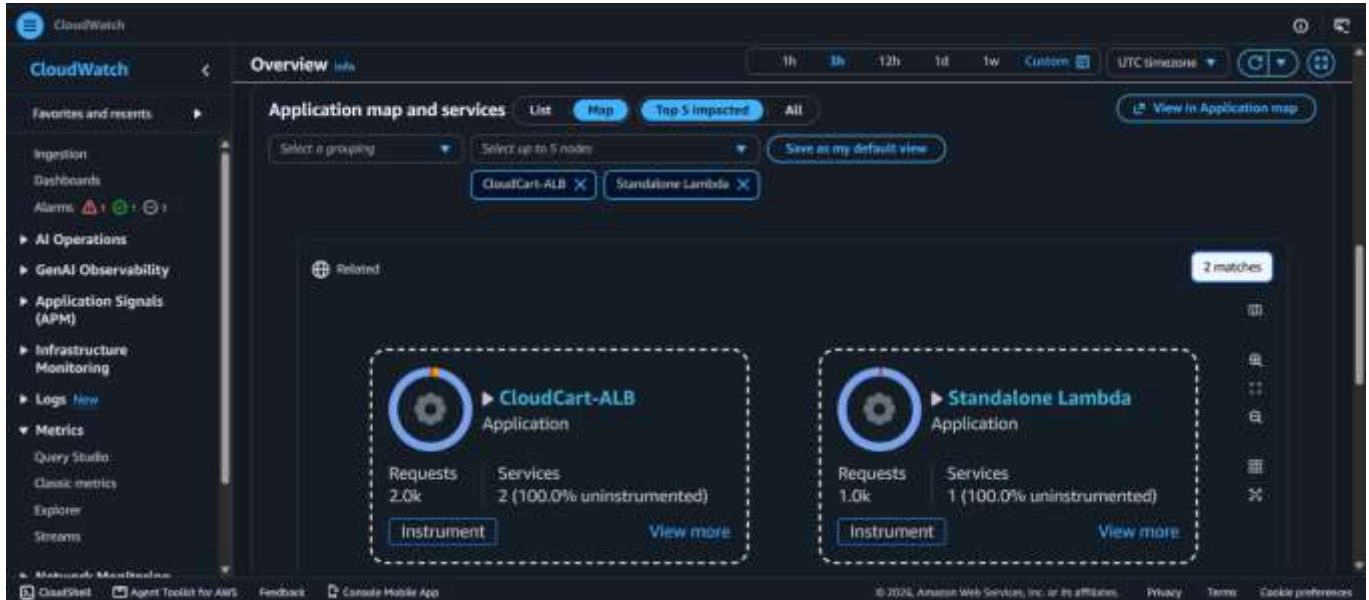
- Creation date:** August 26, 2026, 21:06 (UTC+03:00)
- ARN:** `arn:aws:iam::294427322944:role/service-role/CloudCart-S3-Processor-role-vblgaah0`
- Last activity:** 5 hours ago
- Maximum session duration:** 1 hour

The 'Permissions policies (2)' section shows two attached policies:

Policy name	Type	Attached entities
AWSLambdaBasicExecutionRole-560022d3...	Customer managed	1
CloudCart-S3-Access-Policy	Customer inline	0

To enforce the principle of least privilege, custom **IAM Roles** (CloudCart-EC2-SSM-Role and CloudCart-S3-Processor-role) are configured. These roles securely manage permissions and policies, enabling EC2 instances and Lambda functions to interact with AWS services like Systems Manager and S3 securely without exposing long-term credentials.

3.18 Amazon CloudWatch Configuration & Monitoring



Amazon **CloudWatch** is configured to monitor the operational health and performance metrics of the CloudCart architecture. The dashboard maps core components including the Application Load Balancer (CloudCart-ALB) and serverless functions and tracks resource states, metrics, and alarms to ensure high availability and proactive issue detection.